

CLASSIFYING KNOTTED SURFACES

IZ TECÚM

ABSTRACT. IAS/PCMI lecture notes.

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Date: September 4, 2026.

¹Based on lectures by Anthony Conway [17].

Part 1. Classifying Knotted Surfaces²

These lectures concern locally flat surfaces in simply connected 4-manifolds, where topological surgery and s -cobordism convert geometric classification into algebra. Freedman–Quinn theory [32, 65, 31] yields, among other consequences,

$$\Delta_K(t) = 1 \implies K \subset S^3 \text{ is topologically slice,} \quad \pi_1(S^4 \setminus S) \cong \mathbb{Z} \implies S \subset S^4 \text{ is unknotted.}$$

Lawson’s projective-plane theorem gives the finite-cyclic analogue [49, 27, 46],

$$S \cong \mathbb{RP}^2, \quad \pi_1(S^4 \setminus S) \cong \mathbb{Z}_2 \implies S \subset S^4 \text{ is unknotted.}$$

Lee–Wilczyński developed the finite-cyclic theory for closed orientable surfaces [51, 52, 53, 38], while Boyer treated simply connected exteriors [3]. Surgery later entered knot concordance [34, 23], and recent work has renewed the classification of spheres, discs, nonorientable surfaces, and surfaces with prescribed cyclic knot group [20, 29, 61, 21, 18, 15, 12, 28, 44, 22, 13, 19, 8].

Section 1 fixes the equivalence relation, records the classical classifications, and reviews the topological toolkit. Section 2 treats uniqueness through boundary-compatible equivariant forms and existence through surgery, stable realisation, and branched covers.

1. CLASSIFICATIONS AND THE TOPOLOGICAL TOOLKIT

Throughout, X is a compact, connected, oriented topological 4-manifold, ambient homeomorphisms preserve orientation, and $S \subseteq X$ is compact, locally flat, and properly embedded when boundaries are present. Write

$$X_S = X \setminus \text{int}(v(S)), \quad \pi(S) = \pi_1(X \setminus S),$$

for the exterior and knot group of S , respectively.

1.1. The Meaning of Classification. For good fundamental groups, topological 4-manifolds retain the high-dimensional surgery-theoretic tools needed for classification [32, 65, 31]. The classification problem first requires a precise equivalence relation and purely algebraic data.

Definition 1.1 (Equivalence and isotopy). Two surfaces $S_0, S_1 \subseteq X$ are *equivalent* if some orientation-preserving homeomorphism $\Phi: X \rightarrow X$ satisfies $\Phi(S_0) = S_1$, and are *isotopic* if Φ may be chosen isotopic to id_X .

Definition 1.2 (Classification data). Let $\mathcal{S}(X)$ be a class of surfaces in X , let $\sim$ denote equivalence or isotopy, and let $\mathcal{A}$ be an algebraically defined set. A map $I: \mathcal{S}(X) \rightarrow \mathcal{A}$ is a *classification* when

$$I(S_0) = I(S_1) \iff S_0 \sim S_1, \quad \{I(S) \mid S \in \mathcal{S}(X)\} = \mathcal{A},$$

or equivalently when it induces a bijection

$$\bar{I}: \mathcal{S}(X)/\sim \xrightarrow{\cong} \mathcal{A}.$$

For surfaces with boundary, $\partial S \subseteq \partial X$ is part of the data. One usually fixes genus, boundary, and knot group before classifying; a surface with $\pi(S) \cong \pi$ is a π -*surface*. Concordance gives a different relation studied in [39, 6, 70, 57, 41, 25, 37].

Definition 1.3 (Primary surface invariants). For a closed oriented surface, its homology class is $[S] \in H_2(X)$; use $H_2(X, \partial X)$ for a properly embedded oriented surface and $\mathbb{Z}_2$ -coefficients in the nonorientable case. When X is closed and simply connected, the *divisibility* of an oriented class is

$$d([S]) = \begin{cases} 0, & [S] = 0, \\ \max\{d > 0 \mid [S] = dx \text{ for some } x \in H_2(X)\}, & [S] \neq 0, \end{cases}$$

²Based on lectures by Anthony Conway [17].

while for nonorientable S , set

$$d(S) = \begin{cases} 2, & [S] = 0 \in H_2(X; \mathbb{Z}_2), \\ 1, & [S] \neq 0 \in H_2(X; \mathbb{Z}_2). \end{cases}$$

A class is *primitive* when $d = 1$, and $w \in H_2(X)$ is *characteristic* precisely when

$$w \cdot x \equiv x \cdot x \pmod{2} \quad (x \in H_2(X));$$

otherwise it is *ordinary*. The normal Euler number satisfies

$$e(S) = [S] \cdot [S]$$

for oriented S , and is defined for nonorientable S by intersecting the zero-section with a generic normal push-off [18, Section 2].

The homology class is an isotopy invariant, while its divisibility, type, and Euler number are equivalence invariants. Let $N_{g,e}$ be the D^2 -bundle of Euler number e over the closed genus- g surface of the same orientability type as S , and set

$$Y_{g,e} = \partial N_{g,e}.$$

Lemma 1.4 (Boundary of a surface exterior, [31]). *If X is closed and $S \subset X$ is a closed genus- g surface of Euler number e , then*

$$\partial X_S \cong -Y_{g,e} \quad \text{as oriented 3-manifolds.}$$

Proof. The tubular-neighbourhood theorem identifies the normal disc bundle and its boundary orientation.

$$\begin{aligned} S \subset X \text{ locally flat} &\stackrel{[31, \text{Theorem 9.3}]}{\implies} \overline{\nu(S)} \cong N_{g,e}, & \partial \overline{\nu(S)} &\cong Y_{g,e}, \\ X = X_S \cup_{Y_{g,e}} N_{g,e}, & \partial X_S = -\partial N_{g,e}, & \partial X_S &\cong -Y_{g,e}. \end{aligned}$$

Thus the required oriented boundary identification follows. $\square$

Proposition 1.5 (Homology of sphere exteriors, [17]). *Let X be closed and simply connected, and let $S \subset X$ be a sphere of divisibility $d > 0$. Then*

$$H_k(X_S; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 0, \\ \mathbb{Z}_d, & k = 1, \\ \mathbb{Z}^{b_2(X)-1}, & k = 2, \\ 0, & k = 3, \end{cases} \quad \chi(X_S) = b_2(X).$$

Consequently,

$$\pi(S) \text{ abelian} \implies \pi(S) \cong H_1(X_S) \cong \mathbb{Z}_d.$$

For an arbitrary closed surface, orientable or not, the same conclusion $H_1(X_S) \cong \mathbb{Z}_d$ holds with the divisibility convention of Definition 1.3, where $\mathbb{Z}_0 = \mathbb{Z}$.

Proof. Excision, the Thom isomorphism, and exactness determine the exterior homology.

$$H_k(X, X_S) \cong H_k(\overline{\nu(S)}, \partial \overline{\nu(S)}) \cong H_{k-2}(S^2).$$

$$H_4(X) \xrightarrow{\cong} H_4(X, X_S), \quad H_3(X, X_S) = H_3(X) = 0 \implies H_3(X_S) = 0,$$

$$0 \longrightarrow H_2(X_S) \longrightarrow H_2(X) \xrightarrow{[S] \cdot -} \mathbb{Z} \longrightarrow H_1(X_S) \longrightarrow 0.$$

$$\mathcal{Q}_X \text{ unimodular,} \quad d([S]) = d \implies \text{im}([S] \cdot -) = d\mathbb{Z},$$

$$\ker([S] \cdot -) \cong \mathbb{Z}^{b_2(X)-1}, \quad \text{coker}([S] \cdot -) \cong \mathbb{Z}_d,$$

$$H_0(X_S) \cong \mathbb{Z}, \quad H_1(X_S) \cong \mathbb{Z}_d, \quad H_2(X_S) \cong \mathbb{Z}^{b_2(X)-1},$$

$$\chi(X_S) = \chi(X) - \chi(S^2) = (2 + b_2(X)) - 2 = b_2(X), \quad \pi(S) \text{ abelian} \implies \pi(S) \cong H_1(X_S) \cong \mathbb{Z}_d.$$

For arbitrary S , the Thom sequence uses ordinary coefficients in the orientable case and the normal-orientation local system otherwise.

$$\begin{aligned}
S \text{ orientable} &\implies H_2(X, X_S; \mathbb{Z}) \cong H_0(S; \mathbb{Z}) \cong \mathbb{Z}, \\
H_2(X) &\xrightarrow{[S] \cdot -} \mathbb{Z} \longrightarrow H_1(X_S) \longrightarrow 0, \quad H_1(X_S) \cong \mathbb{Z}/d\mathbb{Z} \cong \mathbb{Z}_d. \\
H_2(X, X_S; \mathbb{Z}) &\cong H_0(S; \mathbb{Z}^{w_1}) \cong \mathbb{Z}_2, \\
H_2(X) &\xrightarrow{[S] \cdot - \bmod 2} \mathbb{Z}_2 \longrightarrow H_1(X_S) \longrightarrow 0, \\
\text{im}(H_2(X) \rightarrow \mathbb{Z}_2) &= \begin{cases} \mathbb{Z}_2, & [S] \neq 0 \in H_2(X; \mathbb{Z}_2), \\ 0, & [S] = 0 \in H_2(X; \mathbb{Z}_2), \end{cases} \quad H_1(X_S) \cong \begin{cases} 0 = \mathbb{Z}_1, & [S] \neq 0, \\ \mathbb{Z}_2, & [S] = 0, \end{cases} \cong \mathbb{Z}_d.
\end{aligned}$$

Thus all asserted homology and knot-group identifications follow. $\square$

The first complete classifications appeared in the work of Freedman, Lawson, Boyer, and Lee–Wilczyński.

Theorem 1.6 (Freedman’s unknotting theorem, [33]). *For a locally flat sphere $S \subset S^4$,*

$$\pi_1(S^4 \setminus S) \cong \mathbb{Z} \iff S \text{ is isotopic to the unknotted sphere.}$$

Thus the isotopy classification of $\mathbb{Z}$ -spheres in S^4 consists of one class.

Theorem 1.7 (Lawson’s unknotting theorem, [49]). *For a locally flat projective plane $S \subset S^4$,*

$$\pi_1(S^4 \setminus S) \cong \mathbb{Z}_2 \iff S \text{ is unknotted.}$$

Theorem 1.8 (Massey’s normal Euler-number theorem, [54]). *If $S \subset S^4$ is closed and nonorientable of genus h , then*

$$e(S) \in \{-2h, -2h+4, \dots, 2h-4, 2h\}.$$

Every value is realised by a connected sum of unknotted projective planes.

The two unknotted projective planes satisfy

$$\pi(S) \cong \mathbb{Z}_2 \xrightarrow{1.7} S \text{ unknotted}, \quad h = 1 \xrightarrow{1.8} e(S) = \pm 2,$$

and their Euler numbers distinguish them.

For a closed simply connected X , write $\sigma(X) \in \mathbb{Z}$ for the signature and $\text{ks}(X) \in \mathbb{Z}_2$ for the Kirby–Siebenmann invariant, normalized by

$$\text{ks}(X) = \begin{cases} 0, & X \#_n (S^2 \times S^2) \text{ is smoothable for some } n \geq 0, \\ 1, & \text{otherwise.} \end{cases}$$

For characteristic $x \in H_2(X)$, the *Kervaire–Milnor condition* is

$$(1) \quad \text{ks}(X) \equiv \frac{\sigma(X) - x \cdot x}{8} \pmod{2}.$$

Kervaire–Milnor proved the smooth necessity of Equation (1), and Freedman–Kirby proved its locally flat analogue [42, 26]. For example,

$$x = 3[\mathbb{CP}^1] \in H_2(\mathbb{CP}^2), \quad \frac{\sigma(\mathbb{CP}^2) - x \cdot x}{8} = -1 \not\equiv 0 = \text{ks}(\mathbb{CP}^2).$$

(1) fails $\xrightarrow{[42, 26]} x$ has no locally flat sphere representative.

Theorem 1.9 (Boyer’s classification, [3]). *Fix a closed simply connected 4-manifold X and $g \geq 0$. Closed oriented genus- g surfaces with simply connected complements satisfy*

$$S_0 \text{ is isotopic to } S_1 \iff [S_0] = [S_1] \in H_2(X).$$

A class $x \in H_2(X)$ is represented by such a surface precisely when

$$x \text{ is primitive and } (g = 0 \text{ and } x \text{ characteristic} \implies (1)).$$

Thus Boyer’s invariant is the primitive homology class, subject only to the Kervaire–Milnor obstruction for characteristic spheres.

Definition 1.10 (Equivariant and pointed intersection forms). Let W have $\pi_1(W) = \pi$ and universal cover $\tilde{W}$. Its equivariant intersection form is

$$\lambda_{\tilde{W}} : H_2(\tilde{W}) \times H_2(\tilde{W}) \longrightarrow \mathbb{Z}[\pi], \quad \lambda_{\tilde{W}}(x, y) = \sum_{g \in \pi} (x \cdot gy) g^{-1}.$$

The sum is finite and

$$\lambda_{\tilde{W}}(y, x) = \overline{\lambda_{\tilde{W}}(x, y)}, \quad \overline{\sum n_g g} = \sum n_g g^{-1}.$$

For $\mathbb{Z}_d = \langle T \mid T^d = 1 \rangle$ and an oriented $\mathbb{Z}_d$ -surface $S \subset X$, let $\Sigma_d(S) \rightarrow X$ be the d -fold branched cover. Its deck-equivariant form is

$$\lambda_{\Sigma_d(S)}(x, y) = \sum_{k=0}^{d-1} Q_{\Sigma_d(S)}(x, T^k y) T^{-k} \in \mathbb{Z}[\mathbb{Z}_d].$$

If $\tilde{S} \subset \Sigma_d(S)$ is the branch set, the *pointed equivariant intersection form* is

$$(\lambda_{\Sigma_d(S)}, [\tilde{S}]);$$

an isometry of pointed forms is a hermitian isometry carrying one distinguished class to the other.

Theorem 1.11 (Lee–Wilczyński uniqueness, [51, 52, 53]). *Fix a closed simply connected 4-manifold X and $d > 0$. For $\mathbb{Z}_d$ -spheres $S_0, S_1 \subset X$,*

$$S_0, S_1 \text{ are equivalent} \iff \begin{cases} S_0, S_1 \text{ are both ordinary or both characteristic,} \\ e(S_0) = e(S_1), \\ (\lambda_{\Sigma_d(S_0)}, [\tilde{S}_0]) \cong (\lambda_{\Sigma_d(S_1)}, [\tilde{S}_1]) \end{cases}$$

If $[S_0] = [S_1] = x$, then S_0 and S_1 are isotopic provided

$$d = 1 \quad \text{or} \quad \left[b_2(X) > 6 \text{ and } b_2(X) > \max_{0 \leq j < d} \left| \sigma(X) - \frac{2j(d-j)}{d^2} x \cdot x \right| \right].$$

Under additional hypotheses on X , homology alone classifies these spheres up to isotopy [52, Theorem 1.1].

Theorem 1.12 (Lee–Wilczyński realisation, [51, 52, 53]). *Let X be a closed simply connected 4-manifold, let $d > 0$, let $g \geq 0$, and let $x \in H_2(X)$. The class x is represented by a genus- g $\mathbb{Z}_d$ -surface if and only if*

$$(2) \quad d(x) = d, \quad b_2(X) + 2g \geq \max_{0 \leq j \leq d} \left| \sigma(X) - \frac{2j(d-j)}{d^2} x \cdot x \right|,$$

and, when $g = 0$ and x is characteristic, x also satisfies Equation (1).

Rochlin established Equation (2) for smooth orientable surfaces when d is a prime power [66]; Freedman–Quinn theory supplies the locally flat setting [31].

Example 1.13 (The unknotted exterior form, [21]). For the unknotted genus- g surface $U \subset S^4$,

$$(H_2(\tilde{X}_U), \lambda_{\tilde{X}_U}) \cong \left(\mathbb{Z}[t^{\pm 1}]^{\oplus 2g}, \begin{pmatrix} 0 & t-1 \\ t^{-1}-1 & 0 \end{pmatrix}^{\oplus g} \right).$$

This follows from a handle decomposition of the exterior [21, Lemma 6.1]; if $g \neq 1, 2$, every $\mathbb{Z}$ -surface $S \subset S^4$ has this equivariant form [21, Theorem 6.4].

For the following double-cover computations, $\mathbb{Z}_-$ denotes the $\mathbb{Z}[\mathbb{Z}_2]$ -module $\mathbb{Z}$ with $Tx = -x$.

Proposition 1.14 (A $\mathbb{Z}_2$ -sphere exterior in $\mathbb{CP}^2$, [17]). *Every $\mathbb{Z}_2$ -sphere $S \subset \mathbb{CP}^2$ satisfies*

$$H_1(\partial \tilde{X}_S) \cong \mathbb{Z}_2, \quad (H_2(\tilde{X}_S), Q_{\tilde{X}_S}) \cong (\mathbb{Z}_-, -2),$$

and

$$\lambda_{\tilde{X}_S}(x, y) = -2(1 - T)xy.$$

Proof. We compute the rank, determinant, sign, and deck action in sequence.

$$[S] = \pm 2[\mathbb{CP}^1] \implies e(S) = 4, \quad \chi(X_S) = \chi(\mathbb{CP}^2) - \chi(S^2) = 1.$$

$$d([S]) = 2 \xrightarrow{1.5} H_1(X_S) \cong \mathbb{Z}_2, \quad H_2(X_S) = H_3(X_S) = 0.$$

$$\pi_1(\partial X_S) \twoheadrightarrow \langle \mu \rangle = \pi_1(X_S) \cong \mathbb{Z}_2 \implies \partial \tilde{X}_S \text{ connected.}$$

$$\chi(\tilde{X}_S) = 2\chi(X_S) = 2, \quad \pi_1(\tilde{X}_S) = 1,$$

$$H_1(\tilde{X}_S) = 0, \quad H_4(\tilde{X}_S) = 0, \quad H_3(\tilde{X}_S) \cong H^1(\tilde{X}_S, \partial \tilde{X}_S) = 0 \implies H_2(\tilde{X}_S) \cong \mathbb{Z}.$$

$$e(S) = 4 \xrightarrow{1.4} \partial X_S \cong -Y_{0,4}, \quad \pi_1(Y_{0,4}) \cong \mathbb{Z}_4 \twoheadrightarrow \mathbb{Z}_2 \implies \partial \tilde{X}_S \cong -Y_{0,2}$$

$$H_1(\partial \tilde{X}_S) \cong \mathbb{Z}_2, \quad H_2(\partial \tilde{X}_S) = 0.$$

$$0 \longrightarrow H_2(\tilde{X}_S) \xrightarrow{Q_{\tilde{X}_S}} H_2(\tilde{X}_S)^* \longrightarrow H_1(\partial \tilde{X}_S) \longrightarrow 0 \implies Q_{\tilde{X}_S} \cong (\varepsilon 2), \quad \varepsilon \in \{\pm 1\}.$$

$$e(S) = 4, \quad \sigma(\mathbb{CP}^2) = 1 \xrightarrow{[17]} \sigma(\Sigma_2(S)) = 2\sigma(\mathbb{CP}^2) - \frac{1}{2}e(S) = 0, \quad [\tilde{S}]^2 = \frac{1}{2}e(S) = 2.$$

$$\Sigma_2(S) = \tilde{X}_S \cup_{Y_{0,2}} N_{0,2}, \quad \sigma(N_{0,2}) = 1, \quad \sigma(\tilde{X}_S) = 0 - 1 = -1 \implies Q_{\tilde{X}_S} \cong (-2).$$

$$H_2(\tilde{X}_S)^T \otimes \mathbb{Q} \cong H_2(X_S; \mathbb{Q}) = 0 \implies Tx = -x \implies H_2(\tilde{X}_S) \cong \mathbb{Z}_-.$$

$$\lambda_{\tilde{X}_S}(x, y) = Q_{\tilde{X}_S}(x, y) + Q_{\tilde{X}_S}(x, Ty)T^{-1} = -2xy + 2xyT = -2(1 - T)xy.$$

Thus the universal exterior has the asserted negative equivariant form. $\square$

Proposition 1.15 (A $\mathbb{Z}_2$ -projective-plane exterior in S^4 , [17]). *Let $S \subset S^4$ be a $\mathbb{Z}_2$ -projective plane of Euler number $e = \pm 2$. Then*

$$H_1(\partial \tilde{X}_S) \cong \mathbb{Z}_4, \quad (H_2(\tilde{X}_S), Q_{\tilde{X}_S}) \cong (\mathbb{Z}_-, -2e),$$

and

$$\lambda_{\tilde{X}_S}(x, y) = -2e(1 - T)xy.$$

Proof. We compute the rank, determinant, sign, and deck action in sequence.

$$e = e(S) = \pm 2, \quad \chi(X_S) = \chi(S^4) - \chi(\mathbb{RP}^2) = 1.$$

$$d(S) = 2 \xrightarrow{1.5} H_1(X_S) \cong \mathbb{Z}_2, \quad H_2(X_S) = H_3(X_S) = 0.$$

$$\pi_1(\partial X_S) \twoheadrightarrow \langle \mu \rangle = \pi_1(X_S) \cong \mathbb{Z}_2 \implies \partial \tilde{X}_S \text{ connected.}$$

$$\chi(\tilde{X}_S) = 2\chi(X_S) = 2, \quad \pi_1(\tilde{X}_S) = 1,$$

$$H_1(\tilde{X}_S) = 0, \quad H_4(\tilde{X}_S) = 0, \quad H_3(\tilde{X}_S) \cong H^1(\tilde{X}_S, \partial \tilde{X}_S) = 0 \implies H_2(\tilde{X}_S) \cong \mathbb{Z}.$$

$$e = \pm 2 \xrightarrow{[17]} \partial \tilde{X}_S \cong -Y_{0,2e} \implies H_1(\partial \tilde{X}_S) \cong \mathbb{Z}_4, \quad H_2(\partial \tilde{X}_S) = 0.$$

$$\begin{aligned}
0 &\longrightarrow H_2(\tilde{X}_S) \xrightarrow{Q_{\tilde{X}_S}} H_2(\tilde{X}_S)^* \longrightarrow H_1(\partial\tilde{X}_S) \longrightarrow 0 \implies Q_{\tilde{X}_S} \cong (\pm 4). \\
H_2(\overline{\nu(\tilde{S})}; \mathbb{Q}) &= 0, \quad e(S) = e \xrightarrow{[17]} \sigma(\tilde{X}_S) = \sigma(\Sigma_2(S)) = 2\sigma(S^4) - \frac{1}{2}e = -\frac{1}{2}e. \\
H_2(\tilde{X}_S) &\cong \mathbb{Z}, \quad \left| \det Q_{\tilde{X}_S} \right| = 4, \quad \text{sgn}(Q_{\tilde{X}_S}) = -\frac{1}{2}e \implies Q_{\tilde{X}_S} \cong (-2e). \\
H_2(\tilde{X}_S)^T \otimes \mathbb{Q} &\cong H_2(X_S; \mathbb{Q}) = 0 \implies Tx = -x \implies H_2(\tilde{X}_S) \cong \mathbb{Z}_-. \\
\lambda_{\tilde{X}_S}(x, y) &= Q_{\tilde{X}_S}(x, y) + Q_{\tilde{X}_S}(x, Ty)T^{-1} = -2exy + 2exyT = -2e(1 - T)xy.
\end{aligned}$$

Thus the Euler number determines the asserted equivariant form. $\square$

Definition 1.16 (Rochlin quadratic form, [55]). Let X be simply connected and let $S \subset X$ be closed, oriented, and characteristic. For $[\gamma] \in H_1(S; \mathbb{Z}_2)$, choose an embedded disk $D \subset X$ with $\partial D = \gamma$ and $\text{int}(D) \cap S$, and let $W(D)$ be the obstruction to extending the surface framing of γ over D . The well-defined quadratic refinement is

$$q_S: H_1(S; \mathbb{Z}_2) \longrightarrow \mathbb{Z}_2, \quad q_S([\gamma]) \equiv S \cdot D + W(D) \pmod{2}.$$

For a symplectic basis $a_1, b_1, \dots, a_g, b_g$ of $H_1(S; \mathbb{Z}_2)$, write

$$\text{Arf}(q_S) = \sum_{i=1}^g q_S(a_i)q_S(b_i) \in \mathbb{Z}_2.$$

Theorem 1.17 (Freedman–Kirby congruence, [26]). *Let X be closed and simply connected. Every closed oriented characteristic surface $S \subset X$ satisfies*

$$\text{Arf}(q_S) \equiv \text{ks}(X) - \frac{\sigma(X) - [S] \cdot [S]}{8} \pmod{2}.$$

In particular, if X is spin, then

$$(3) \quad \text{ks}(X) \equiv \frac{\sigma(X)}{8} \pmod{2}.$$

The nonorientable analogue is the $\mathbb{Z}_4$ -valued Guillou–Marin form [36]; boundary versions appear in [40].

1.2. Tools of the Trade. An h -cobordism $(W; M_0, M_1)$ has both inclusions $M_i \hookrightarrow W$ as homotopy equivalences; it is an s -cobordism when they are simple homotopy equivalences [50]. For a homeomorphism $f: \partial M_0 \rightarrow \partial M_1$, the notation $(W; M_0, M_1)$ rel. f records

$$\partial W = M_0 \cup_{\text{id}} (\partial M_0 \times [0, 1]) \cup_f M_1.$$

The classical theorem holds in dimensions at least six [59, 48]; Freedman–Quinn theory supplies the five-dimensional theorem when the fundamental group is good [32, 65, 31]. Finite and solvable groups are good; in particular, abelian groups are good [35, 1].

Theorem 1.18 (Topological s -cobordism rel. boundary, [59, 48, 32, 65, 31]). *Let $(W; M_0, M_1)$ be an $(n+1)$ -dimensional s -cobordism rel. f . If $n \geq 5$, or if $n = 4$ and $\pi_1(W)$ is good, then*

$$W \cong M_0 \times [0, 1] \quad \text{rel.} \quad M_0 \cup (\partial M_0 \times [0, 1]).$$

Corollary 1.19 (Relative s -cobordism extension, [17]). *Under the hypotheses of Theorem 1.18, the boundary homeomorphism f extends to a homeomorphism $M_0 \rightarrow M_1$.*

Proof. We restrict the relative product structure to its far end.

$$\begin{array}{ccc}
 P: M_0 \times [0, 1] & \xrightarrow{\cong} & W \quad \text{rel.} \quad M_0 \cup (\partial M_0 \times [0, 1]) \\
 \downarrow G & & \downarrow P \\
 M_1 & \xrightarrow{\quad} & W
 \end{array}
 \quad G = P|_{M_0 \times \{1\}}: M_0 \xrightarrow{\cong} M_1, \quad G|_{\partial M_0} = f.$$

Thus G is the required extension. $\square$

Classical surgery was developed by Browder, Novikov, Sullivan, and Wall [5, 4, 60, 69, 71]; Kreck's modified surgery starts from a controlled normal cobordism rather than a prescribed homotopy equivalence [45, 47]. We use the following four-dimensional consequence.

Theorem 1.20 (Surgery to an s -cobordism, [32, 9]). *Let X_0, X_1 be compact, connected, oriented topological 4-manifolds with*

$$\pi = \pi_1(X_0) \cong \pi_1(X_1), \quad \pi \in \{1, \text{finite cyclic, solvable Baumslag–Solitar}\}.$$

When $\partial X_i \neq \emptyset$, assume that

$$\pi_1(\partial X_i) \longrightarrow \pi_1(X_i) \quad \text{is surjective} \quad (i = 0, 1).$$

If

$$\text{ks}(X_0, \partial X_0) = \text{ks}(X_1, \partial X_1)$$

and $f: \partial X_0 \rightarrow \partial X_1$ extends to a homotopy equivalence $X_0 \rightarrow X_1$, then X_0, X_1 are s -cobordant rel. f .

The restrictions on π are substantive, since beyond these groups the surgery obstruction and its geometric realisation are generally much harder to control [9].

Theorem 1.21 (Freedman's classification, [32]). *Closed, oriented, simply connected topological 4-manifolds X_0, X_1 are orientation-preservingly homeomorphic if and only if*

$$Q_{X_0} \cong Q_{X_1}, \quad \text{ks}(X_0) = \text{ks}(X_1).$$

Every unimodular symmetric integral form Q is realised by a closed simply-connected topological 4-manifold. If Q is even,

$$\text{ks}(X) \equiv \frac{\text{sgn}(Q)}{8} \pmod{2},$$

whereas either Kirby–Siebenmann invariant occurs when Q is odd.

Proof. We pass from form data to an s -cobordism and realise an arbitrary form by a capped framed-link trace.

$$\begin{aligned}
 & \left. \begin{array}{l} Q_{X_0} \cong Q_{X_1}, \\ \text{ks}(X_0) = \text{ks}(X_1) \end{array} \right\} \xrightarrow{[72, 58, 56]} X_0 \simeq X_1 \xrightarrow{1.20} X_0 \sim_s X_1 \xrightarrow{1.18} X_0 \cong X_1. \\
 & X_0 \cong X_1 \implies Q_{X_0} \cong Q_{X_1}, \quad \text{ks}(X_0) = \text{ks}(X_1). \\
 & Q \text{ unimodular} \implies \exists L \subset S^3. \\
 & \text{lk}(L) = Q, \quad H_1(\partial X(L)) = 0, \quad X(L) = B^4 \cup_L \{\text{framed 2-handles}\}. \\
 & H_1(\partial X(L)) = 0 \xrightarrow{[32]} \exists C, \quad \partial C \cong -\partial X(L), \quad \pi_1(C) = 1, \quad H_*(C) \cong H_*(\text{pt}). \\
 & L \subseteq S^3 \longmapsto X(L) = B^4 \cup_L \{\text{framed 2-handles}\} \longmapsto X = X(L) \cup_{\partial X(L)} C. \\
 & \pi_1(X(L)) = 1, \quad Q_{X(L)} \cong Q, \quad X = X(L) \cup_{\partial} C \implies \pi_1(X) = 1, \quad Q_X \cong Q. \\
 & Q \text{ even} \xrightarrow{(3)} \text{ks}(X) \equiv \frac{\sigma(X)}{8} = \frac{\text{sgn}(Q)}{8} \pmod{2}.
 \end{aligned}$$

$$Q \text{ odd} \xrightarrow{[31, 68, 43]} \begin{cases} Q_{X^*} \cong Q_X \cong Q, \\ \text{ks}(X^*) = \text{ks}(X) + 1 \end{cases} \implies \text{ks} \in \mathbb{Z}_2 \text{ is arbitrary.}$$

This proves both classification and realisation. $\square$

In particular, the intersection form alone classifies closed simply-connected spin 4-manifolds, since Equation (3) recovers ks from its signature.

The disc-embedding theorem replaces the Whitney trick for good groups [31, 1]. Write $\tilde{\mu}_V$ for the reduced self-intersection invariant on $\pi_2(V) \cong H_2(\tilde{V})$; the sphere-embedding consequence converts a hyperbolic quadratic summand into a connected summand [63].

Theorem 1.22 (Hyperbolic splitting, [31, 1, 63]). *Let V be a 4-manifold with good fundamental group π , put $\Lambda = \mathbb{Z}[\pi]$, and suppose the direct summand*

$$H = \bigoplus_{i=1}^k (\Lambda a_i \oplus \Lambda b_i) \subseteq \pi_2(V) \cong H_2(\tilde{V})$$

satisfies

$$\lambda_{\tilde{V}}(a_i, a_j) = 0, \quad \lambda_{\tilde{V}}(b_i, b_j) = 0, \quad \lambda_{\tilde{V}}(a_i, b_j) = \delta_{ij}, \quad \tilde{\mu}_V(a_i) = \tilde{\mu}_V(b_i) = 0.$$

There is a 4-manifold W and a homeomorphism

$$V \cong W \#_{i=1}^k (S^2 \times S^2).$$

The induced map carries each (a_i, b_i) to the standard hyperbolic basis of the corresponding $S^2 \times S^2$ factor.

We now apply these tools to closed surfaces in a fixed closed simply-connected 4-manifold X . By Lemma 1.4, a genus- g , Euler-number- e surface has oriented exterior boundary $-Y_{g,e}$.

Proposition 1.23 (Equivalence from an exterior homotopy equivalence, [9, 17]). *Let $S_0, S_1 \subset X$ have common knot group π , where π is trivial, finite cyclic, or a solvable Baumslag–Solitar group, and assume the peripheral maps $\pi_1(\partial X_{S_i}) \twoheadrightarrow \pi_1(X_{S_i})$ are surjective. Fix an orientation-preserving homeomorphism*

$$H: (\bar{v}(S_0), S_0) \xrightarrow{\cong} (\bar{v}(S_1), S_1).$$

If $H|_{\partial}$ extends to a homotopy equivalence $h: X_{S_0} \rightarrow X_{S_1}$, then S_0, S_1 are equivalent.

Proof. Kirby–Siebenmann additivity [30] matches the exterior invariants, after which surgery fixes the prescribed boundary map.

$$\text{ks}(X_{S_0}, \partial X_{S_0}) = \text{ks}(X) - \text{ks}(\bar{v}(S_0), \partial \bar{v}(S_0)) = \text{ks}(X) - \text{ks}(\bar{v}(S_1), \partial \bar{v}(S_1)) = \text{ks}(X_{S_1}, \partial X_{S_1}).$$

$$h: X_{S_0} \simeq X_{S_1} \text{ rel. } H|_{\partial} \xrightarrow{1.20} X_{S_0} \sim_s X_{S_1} \text{ rel. } H|_{\partial} \xrightarrow{1.18, 1.19} \hat{H}: X_{S_0} \xrightarrow{\cong} X_{S_1}, \quad \hat{H}|_{\partial} = H|_{\partial}.$$

$$\begin{array}{ccc} \partial X_{S_0} & \xrightarrow{H|_{\partial}} & \partial X_{S_1} \\ \downarrow & & \downarrow \\ X_{S_0} & \xrightarrow{\hat{H}} & X_{S_1} \\ \downarrow & & \downarrow \\ \bar{v}(S_0) & \xrightarrow{H} & \bar{v}(S_1) \\ \hat{H} \cup_{H|_{\partial}} H & \xrightarrow{\cong} & (X, S_0) \xrightarrow{\cong} (X, S_1) \end{array}$$

Thus S_0 and S_1 are equivalent. $\square$

Proposition 1.24 (Realisation from a circle-bundle filling, [17]). *Let W be a compact, connected, oriented topological 4-manifold. Suppose $\partial W \cong -Y_{g,e}$, where $Y_{g,e} = \partial N_{g,e}$, and assume*

$$\text{ks}(W, \partial W) = \text{ks}(X), \quad \pi_1(W) \cong \pi, \quad \widehat{W} = W \cup_{Y_{g,e}} N_{g,e} \text{ is simply connected}, \quad Q_{\widehat{W}} \cong Q_X.$$

The manifold X contains a closed genus- g π -surface of Euler number e . If the zero-section $\Sigma_g \subset N_{g,e}$ is orientable and there is an isometry

$$\varphi: Q_{\widehat{W}} \xrightarrow{\cong} Q_X, \quad \varphi([\Sigma_g]) = x \in H_2(X),$$

the surface may be chosen to represent x .

Proof. Kirby–Siebenmann additivity [30] and Freedman’s pointed classification [32] identify the capped filling with X and transport its zero-section.

$$\text{ks}(\widehat{W}) = \text{ks}(W, \partial W) + \text{ks}(N_{g,e}, \partial N_{g,e}) = \text{ks}(X) + 0 = \text{ks}(X).$$

$$\left. \begin{array}{l} \pi_1(\widehat{W}) = 1, \\ Q_{\widehat{W}} \cong Q_X, \\ \text{ks}(\widehat{W}) = \text{ks}(X) \end{array} \right\} \xrightarrow{1.21} \Phi: \widehat{W} \xrightarrow{\cong} X.$$

$$\begin{array}{ccccc} N_{g,e} & \hookrightarrow & \widehat{W} = W \cup_{Y_{g,e}} N_{g,e} & \xrightarrow{\Phi} & X \\ \uparrow & & & & \uparrow \\ \Sigma_g & \xrightarrow{S=\Phi(\Sigma_g)} & & & S. \end{array}$$

$$S \cong \Sigma_g, \quad e(S) = e, \quad X_S \cong W, \quad \pi_1(X \setminus S) \cong \pi_1(W) \cong \pi.$$

$$\Sigma_g \text{ orientable}, \quad \varphi([\Sigma_g]) = x \xrightarrow{[32]} \Phi \text{ may be chosen with } \Phi_* = \varphi, \quad [S] = \Phi_*[\Sigma_g] = x.$$

Thus $S \subset X$ has the required genus, Euler number, and knot group, and represents x in the pointed case. $\square$

The group and simple-connectivity conditions in the preceding proposition are often accessible; matching the capped intersection form is the main obstruction. Hyperbolic splitting supplies the first stable cancellation principle.

Proposition 1.25 (Stable cancellation for spheres, [17]). *Let $x \in H_2(X)$. If*

$$x \oplus 0 \in H_2(X \#_k (S^2 \times S^2))$$

is represented by a sphere with trivial knot group, then x is represented in X by a sphere with trivial knot group.

Proof. The stabilising hyperbolic basis lifts to the simply connected exterior, splits geometrically, and leaves a capped filling of X .

$$X_k = X \#_k (S^2 \times S^2), \quad [S] = x \oplus 0, \quad e = x \cdot x.$$

$$Q_{X_k}|_{\oplus_i (\mathbb{Z}\bar{a}_i \oplus \mathbb{Z}\bar{b}_i)} \cong \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^{\oplus k}, \quad [S] \cdot \bar{a}_i = [S] \cdot \bar{b}_i = 0.$$

$$H_2((X_k)_S) \xrightarrow{j_*} [S]^\perp, \quad j_*(a_i) = \bar{a}_i, \quad j_*(b_i) = \bar{b}_i.$$

$$\bigoplus_{i=1}^k (\mathbb{Z}a_i \oplus \mathbb{Z}b_i) \subseteq_{\text{dir}} H_2((X_k)_S), \quad Q_{(X_k)_S}|_{\oplus_i (\mathbb{Z}a_i \oplus \mathbb{Z}b_i)} \cong \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^{\oplus k},$$

$$\pi_1((X_k)_S) = 1 \implies \tilde{\mu}_{(X_k)_S}(a_i) = \tilde{\mu}_{(X_k)_S}(b_i) = 0 \xrightarrow{1.22} (X_k)_S \cong W \#_k (S^2 \times S^2).$$

$$\begin{array}{ccc}
(X_k)_S & \xrightarrow{\cong} & W \#_k (S^2 \times S^2) \\
\downarrow \cup_{Y_{0,e}} N_{0,e} & & \downarrow \cup_{Y_{0,e}} N_{0,e} \\
X \#_k (S^2 \times S^2) & \xrightarrow{\cong} & \widehat{W} \#_k (S^2 \times S^2), \quad \widehat{W} = W \cup_{Y_{0,e}} N_{0,e}.
\end{array}$$

$\partial W \cong -Y_{0,e}, \quad \pi_1(W) = 1, \quad (Q_{\widehat{W}}, [\Sigma_0]) \cong (Q_X, x), \quad \Sigma_0 \subset N_{0,e}.$
 $\text{ks}(W, \partial W) = \text{ks}((X_k)_S, \partial(X_k)_S) = \text{ks}(X_k) = \text{ks}(X).$
 $\pi_1(\widehat{W}) \cong \pi_1(W) *_{\pi_1(Y_{0,e})} \pi_1(N_{0,e}) = 1 *_{\pi_1(Y_{0,e})} 1 = 1.$
 $\left. \begin{array}{l} \partial W \cong -Y_{0,e}, \quad \pi_1(W) = 1, \quad \pi_1(\widehat{W}) = 1, \\ \text{ks}(W, \partial W) = \text{ks}(X), \quad (Q_{\widehat{W}}, [\Sigma_0]) \cong (Q_X, x) \end{array} \right\} \xrightarrow{1.24} \exists S' \subset X.$
 $S' \cong S^2, \quad [S'] = x, \quad \pi_1(X \setminus S') = 1.$

Thus the stabilised sphere descends to the required sphere in X . $\square$

This cancellation principle converts stable surface results into unstable ones; the same strategy underlies [51, 52, 53, 19, 8].

2. UNIQUENESS AND EXISTENCE

Throughout, X is closed and simply connected, $S \subset X$ is a closed $\mathbb{Z}_d$ -surface, and $\Lambda_d = \mathbb{Z}[\mathbb{Z}_d]$. We use the exterior, cover, and equivariant form of Definition 1.10; boundary maps denote their coefficient-compatible lifts.

2.1. Uniqueness Results. Equivalent surfaces have equal normal Euler number, the same ordinary-characteristic type, and isometric equivariant forms [52, 11]. The boundary determines whether an isometry extends across the normal bundle.

Definition 2.1 (Boundary isomorphism, [17]). For an isometry

$$F: \lambda_{\widetilde{X}_{S_0}} \xrightarrow{\cong} \lambda_{\widetilde{X}_{S_1}},$$

its *boundary isomorphism* is the dashed map in

$$\begin{array}{ccccccc}
H_2(\widetilde{X}_{S_0}) & \longrightarrow & \overbrace{H_2(\widetilde{X}_{S_0}, \partial \widetilde{X}_{S_0})}^{\cong H_2(\widetilde{X}_{S_0})^*} & \longrightarrow & H_1(\partial \widetilde{X}_{S_0}) & \longrightarrow & 0 \\
\downarrow F & & \downarrow (F^*)^{-1} & & \downarrow \partial F & & \\
H_2(\widetilde{X}_{S_1}) & \longrightarrow & \overbrace{H_2(\widetilde{X}_{S_1}, \partial \widetilde{X}_{S_1})}^{\cong H_2(\widetilde{X}_{S_1})^*} & \longrightarrow & H_1(\partial \widetilde{X}_{S_1}) & \longrightarrow & 0,
\end{array}$$

where

$$H_2(\widetilde{X}_{S_i})^* = \overline{\text{Hom}_{\Lambda_d}(H_2(\widetilde{X}_{S_i}), \Lambda_d)}.$$

The bar twists by $T \mapsto T^{-1}$, and the middle map is dual to F .

Lemma 2.2 (Equivariant duality, [17]). *For every $\mathbb{Z}_d$ -surface $S \subset X$,*

$$H_2(\widetilde{X}_S, \partial \widetilde{X}_S) \cong \overline{\text{Hom}_{\Lambda_d}(H_2(\widetilde{X}_S), \Lambda_d)}.$$

Proof. The equivariant universal-coefficient sequence collapses in degree two.

$$\begin{aligned} H_2(\tilde{X}_S, \partial\tilde{X}_S) &\cong H^2(X_S; \Lambda_d), \\ H_1(\tilde{X}_S) &= 0, \quad H_0(\tilde{X}_S) \cong \mathbb{Z}, \quad \text{Ext}_{\Lambda_d}^p(\mathbb{Z}, \Lambda_d) \cong H^p(\mathbb{Z}_d; \Lambda_d) = 0 \quad (p > 0), \\ H^2(X_S; \Lambda_d) &\cong \overline{\text{Hom}_{\Lambda_d}(H_2(\tilde{X}_S), \Lambda_d)}. \end{aligned}$$

The composite is the asserted duality isomorphism. $\square$

For a bundle isomorphism $H: \bar{v}(S_0) \rightarrow \bar{v}(S_1)$, call (F, H) *compatible* when

$$\partial F = (H|_{\partial})_*: H_1(\partial\tilde{X}_{S_0}) \longrightarrow H_1(\partial\tilde{X}_{S_1}).$$

Lemma 2.3 (Compatible boundary data, [31, 17]). *Equivalent $\mathbb{Z}_d$ -surfaces admit a compatible pair*

$$F: \lambda_{\tilde{X}_{S_0}} \xrightarrow{\cong} \lambda_{\tilde{X}_{S_1}}, \quad H: \bar{v}(S_0) \xrightarrow{\cong} \bar{v}(S_1).$$

Proof. Restrict an equivalence and lift its exterior map. After the tubular neighbourhood isotopy of [31, Theorem 9.3],

$$\Phi|_{\bar{v}(S_0)} = H, \quad \Phi|_{X_{S_0}} = h, \quad h|_{\partial} = H|_{\partial}, \quad F = \tilde{h}_*.$$

Naturality and Lemma 2.2 give

$$\partial F = (\tilde{h}|_{\partial})_* = (H|_{\partial})_*.$$

Thus (F, H) is compatible. $\square$

We use the relative first k -invariants of [9, 10]; for characteristic S , let q_S denote the Rochlin form of Definition 1.16.

Theorem 2.4 (Boundary-compatible uniqueness, [9, 10, 11]). *Let $S_0, S_1 \subset X$ be closed orientable $\mathbb{Z}_d$ -surfaces with equal Euler number, either both ordinary or both characteristic. Then*

$$(X, S_0) \cong (X, S_1) \iff \begin{cases} \text{there is a compatible pair } (F, H), \\ (F, H) \text{ preserves the relative first } k\text{-invariants,} \\ q_{S_1} \circ (H|_{S_0})_* = q_{S_0} \text{ in the characteristic case.} \end{cases}$$

The k -condition is automatic for $d = 1$, absent for spheres, and absent in the infinite-cyclic analogue; the quadratic condition is absent for spheres. The nonorientable statement holds unless X is spin and $d = 1$, and a boundary version also holds.

Proof. Naturality proves necessity, and relative surgery proves sufficiency. By Lemma 2.3, an equivalence preserves the compatible pair, relative k -invariants, and characteristic Rochlin form.

Conversely, [9, 10] reduces the extension of $H|_{\partial}$ to the stated data, the spin obstruction, and

$$\begin{array}{ccccccc} H_2(\partial\tilde{X}_{S_0}) & \longrightarrow & H_2(\tilde{X}_{S_0}) & \longrightarrow & \overbrace{H_2(\tilde{X}_{S_0}, \partial\tilde{X}_{S_0})}^{\cong H_2(\tilde{X}_{S_0})^*} & \longrightarrow & H_1(\partial\tilde{X}_{S_0}) \longrightarrow 0 \\ \downarrow (H|_{\partial})_* & & \downarrow F & & \downarrow (F^*)^{-1} & & \downarrow (H|_{\partial})_* \\ H_2(\partial\tilde{X}_{S_1}) & \longrightarrow & H_2(\tilde{X}_{S_1}) & \longrightarrow & \overbrace{H_2(\tilde{X}_{S_1}, \partial\tilde{X}_{S_1})}^{\cong H_2(\tilde{X}_{S_1})^*} & \longrightarrow & H_1(\partial\tilde{X}_{S_1}) \longrightarrow 0. \end{array}$$

Definition 2.1 gives the right square, and boundary intersections give the left square [9, 11]. The spin obstruction is

$$w_2(X_{S_i}) = 0 \iff (w_2(X) = 0) \text{ or } (S_i \text{ is characteristic}),$$

$$\begin{aligned}
w_2(X) = 0 &\implies \text{choose a compatible pair with spin double,} \\
w_2(X_{S_i}) \neq 0, w_2(\tilde{X}_{S_i}) = 0 &\implies d \text{ is even and every } H \text{ gives a spin lifted double,} \\
S_i \text{ characteristic} &\implies X_{S_0} \cup_{H|_\partial} X_{S_1} \text{ is spin} \xLeftrightarrow{[111]} q_{S_1} \circ (H|_{S_0})_* = q_{S_0}.
\end{aligned}$$

Thus the prescribed boundary map extends, and

$$H|_\partial \subset h: X_{S_0} \simeq X_{S_1} \xrightarrow{1,23} (X, S_0) \cong (X, S_1).$$

This proves boundary-compatible uniqueness. $\square$

Remark 2.5 (Special cases of uniqueness). The theorem recovers Boyer, Lee–Wilczyński, and the infinite-cyclic classification [3, 52, 21, 11].

$$\begin{aligned}
&\left. \begin{array}{l} [S_0] = [S_1], \quad g(S_0) = g(S_1), \\ \pi_1(X_{S_0}) = \pi_1(X_{S_1}) = 1 \end{array} \right\} \xrightarrow{1.9} S_0 \simeq_{\text{isotopy}} S_1, \\
(X, S_0) \cong (X, S_1) &\xLeftrightarrow{[52, 11]} (\lambda_{\Sigma_d(S_0)}, [\tilde{S}_0]) \cong (\lambda_{\Sigma_d(S_1)}, [\tilde{S}_1]) \quad \text{by a pointed isometry,} \\
S \subset S^4, \quad \pi_1(S^4 \setminus S) \cong \mathbb{Z}, \quad g(S) \notin \{1, 2\} &\xrightarrow{[21, \text{Theorem 1.1}]} S \text{ is unknotted.}
\end{aligned}$$

For a nonorientable $\mathbb{Z}_2$ -surface of genus h in S^4 ,

$$|e(S)| \neq 2h \text{ or } (h \leq 5, |e(S)| = 2h) \xrightarrow{[49, 27, 46, 18, 62]} S \text{ is unknotted.}$$

A knot in S^3 bounds at most one $\mathbb{Z}_2$ -Möbius band in B^4 , up to equivalence [11, Theorem 1.1].

Example 2.6 ($\mathbb{Z}_2$ -spheres in $\mathbb{CP}^2$, [52, 21, 13, 11]). Any two $\mathbb{Z}_2$ -spheres in $\mathbb{CP}^2$ are equivalent, and homologous ones are isotopic [13, Theorem 1.1]. By Proposition 1.14,

$$e(S_i) = 4, \quad H_1(\partial \tilde{X}_{S_i}) \cong \mathbb{Z}_2, \quad \text{Aut}(\mathbb{Z}_2) = 1, \quad \lambda_{\tilde{X}_{S_i}} \cong (\mathbb{Z}_-, -2(1-T)).$$

Compatibility is automatic, hence

$$(F, H) \xrightarrow{2.4} (\mathbb{CP}^2, S_0) \cong (\mathbb{CP}^2, S_1).$$

Moreover,

$$x = d[\mathbb{CP}^1] \text{ has a sphere representative with abelian knot group } \iff |d| \in \{0, 1, 2\},$$

and these give precisely five isotopy classes [13, Theorem 1.2].

Corollary 2.7 (Lawson’s unknotting criterion, [49, 11]). *If $S \subset S^4$ is a locally flat projective plane with $\pi_1(S^4 \setminus S) \cong \mathbb{Z}_2$, then S is unknotted.*

Proof. We compare S with the standard projective plane of equal Euler number. By Theorem 1.8 and Proposition 1.15,

$$\begin{aligned}
e(S) = e(U) = e \in \{-2, 2\}, \quad H_1(\partial \tilde{X}_S) \cong \mathbb{Z}_4 \cong H_1(\partial \tilde{X}_U), \\
\lambda_{\tilde{X}_S} \cong (\mathbb{Z}_-, -2e(1-T)) \cong \lambda_{\tilde{X}_U}.
\end{aligned}$$

Choose an isometry F and a normal-bundle isomorphism H . Since

$$\text{Aut}(\mathbb{Z}_4) = \{\pm 1\},$$

one of F and $-F$ satisfies $\partial F = (H|_\partial)_*$. The relative k -obstruction vanishes [11], so

$$(F, H) \xrightarrow{2.4} (S^4, S) \cong (S^4, U) \implies S \text{ is unknotted.}$$

This proves Lawson’s criterion. $\square$

For properly embedded surfaces see [21, 7, 64, 18, 11], for discs see [20, 15], for spheres in knot traces see [61], and for immersed $\mathbb{Z}$ -surfaces see [12]. With nonempty boundary the equivariant form need not be complete [7].

2.2. Existence Results. Let T generate $\mathbb{Z}_d$, and set

$$\mathcal{H}_{\Lambda_d} = \left(\Lambda_d a \oplus \Lambda_d b, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tilde{\mu}(a) = \tilde{\mu}(b) = 0 \right).$$

By Lemma 1.4, existence reduces to filling $-Y_{g,e}$. We use stable cancellation [51, 52, 53, 8], cobordism and capping [3, 22, 14, 16], and surgery on Poincaré fillings for aspherical exteriors [32, 31, 34, 23, 29]. Boundary versions occur in [19, 22, 12], while [44] gives another construction.

Lemma 2.8 (Equivariant destabilisation, [51, 52, 31]). *Let X be closed and simply connected, $x \in H_2(X)$, and $X_k = X \#_k (S^2 \times S^2)$. Suppose a $\mathbb{Z}_d$ -sphere $S \subset X_k$ represents $x \oplus 0$ and*

$$(\lambda_{\Sigma_d(S)}, [\tilde{S}]) \cong (\lambda', z') \oplus (\mathcal{H}_{\Lambda_d}^{\oplus k}, 0), \quad (\lambda', z') \otimes_{\Lambda_d} \mathbb{Z} \cong (Q_X, x),$$

with vanishing reduced self-intersections on the displayed hyperbolic bases. Then x is represented in X by a $\mathbb{Z}_d$ -sphere.

Proof. The pointed splitting makes each stabilising hyperbolic pair geometrically removable.

$$\begin{aligned} \mathcal{H}_{\Lambda_d}^{\oplus k} \subseteq H_2(\widetilde{(X_k)_S}) &\cong \pi_2((X_k)_S), & \tilde{\mu}(a_i) = \tilde{\mu}(b_i) = 0 &\xrightarrow{1.22} (X_k)_S \cong W \#_k (S^2 \times S^2), \\ \pi_1(W) &\cong \mathbb{Z}_d, & \partial W &\cong -Y_{0,x,x}. \end{aligned}$$

Capping commutes with cancellation.

$$\begin{array}{ccc} (X_k)_S & \xrightarrow{\cong} & W \#_k (S^2 \times S^2) \\ \downarrow \cup N_{0,x,x} & & \downarrow \cup N_{0,x,x} \\ X \#_k (S^2 \times S^2) & \xrightarrow{\cong} & \widehat{W} \#_k (S^2 \times S^2), \quad \widehat{W} := W \cup N_{0,x,x}. \end{array}$$

For the cap zero-section $\Sigma \subset \widehat{W}$, van Kampen, transfer, Mayer–Vietoris, and relative additivity give

$$\pi_1(\widehat{W}) = 1, \quad (Q_{\widehat{W}}, [\Sigma]) \cong (\lambda', z') \otimes_{\Lambda_d} \mathbb{Z} \cong (Q_X, x), \quad \text{ks}(\widehat{W}) = \text{ks}(X).$$

Proposition 1.24 carries Σ to the required sphere in X . □

Theorem 2.9 (Lee–Wilczyński sphere realisation, [51, 52, 53]). *Let X be closed and simply connected, $d > 0$, and $x \in H_2(X)$. The class x is represented by a $\mathbb{Z}_d$ -sphere if and only if*

$$d(x) = d, \quad b_2(X) \geq \max_{0 \leq j \leq d} \left| \sigma(X) - \frac{2j(d-j)}{d^2} x \cdot x \right|,$$

and, when x is characteristic,

$$\text{ks}(X) \equiv \frac{\sigma(X) - x \cdot x}{8} \pmod{2}.$$

Proof. Necessity follows from exterior homology, Rochlin’s inequality, and the Freedman–Kirby congruence.

$$x = [S], \quad \pi_1(X_S) \cong \mathbb{Z}_d \xrightarrow{1.5} d(x) = d,$$

$$x = [S] \xrightarrow{[66, 52]} (2), \quad x \text{ characteristic} \xrightarrow{1.17} (1).$$

For sufficiency, represent x by a connected oriented surface $F \subset X$. If $\gamma = \partial D \subset F$, put

$$o(D, F) = W(D) + D \cdot F \pmod{2}, \quad o(D, F) = 0 \iff \text{the disc and surface framings agree.}$$

After stabilisation an odd dual corrects $o(D, F)$ when x is ordinary. When x is characteristic,

$$o(D, F) = q_F([\gamma]), \quad \text{Arf}(q_F) \equiv \text{ks}(X) - \frac{\sigma(X) - x \cdot x}{8} \xleftrightarrow{(1)} 0 \pmod{2},$$

$$\text{Arf}(q_F) = 0 \xrightarrow{[51, 67]} x \oplus 0 = [S_0] \in H_2(X \#_r (S^2 \times S^2)).$$

Surgeries on push-offs of commutators then give

$$d(x) = d \xrightarrow{1.5} H_1((X \#_r (S^2 \times S^2))_{S_0}) \cong \mathbb{Z}_d \xrightarrow{[51, 52]} \pi_1((X \#_k (S^2 \times S^2))_S) \cong \mathbb{Z}_d.$$

The rank–signature inequality gives

$$(2) \xrightarrow{[51, 52, 53]} (\lambda_{\Sigma_d(S)}, [\tilde{S}]) \cong (\lambda', z') \oplus (\mathcal{H}_{\Lambda_d}^{\oplus k}, 0), \quad (\lambda', z') \otimes_{\Lambda_d} \mathbb{Z} \cong (Q_X, x),$$

with $\tilde{\mu}(a_i) = \tilde{\mu}(b_i) = 0$. Lemma 2.8 removes the stabilisation, proving sufficiency. $\square$

Lee–Wilczyński theory realises homology classes, while prescribed equivariant forms are known in fewer cases [22, 8]. Write $\cong_s$ for stable isometry.

Lemma 2.10 (Fixed-boundary stable realisation, [8]). *Let Y be a closed 3-manifold, $\varphi: \pi_1(Y) \rightarrow \pi$ have finite target, and $\Lambda = \mathbb{Z}[\pi]$. A nonsingular Λ -hermitian form is realised by a filling W satisfying*

$$\partial W \cong -Y, \quad \pi_1(W) \cong \pi, \quad (\pi_1(Y) \rightarrow \pi_1(W)) = \varphi$$

if and only if it is stably realised by such a filling.

Proof. Only cancellation requires proof. The fixed-boundary cancellation theorem removes the hyperbolic summands while preserving Y and φ [8, Proposition 2.1], giving exact realisation. $\square$

Theorem 2.11 (A nonsmoothable surface in the four-sphere, [8]). *There is a nonorientable genus-9 $\mathbb{Z}_2$ -surface $S \subset S^4$ with Euler number -18 and*

$$Q_{\Sigma_2(S)} \cong E_8 \oplus (1).$$

In particular, S is not topologically equivalent to a smooth surface.

Proof. Fixed-boundary realisation constructs the exterior, and Donaldson’s theorem obstructs smoothability. Put

$$b = E_8 \oplus (1), \quad h = \text{rk}(b) = 9, \quad e = -2h = -18, \quad H_b \cong \mathbb{Z}_- \oplus \Lambda_2^8.$$

For the unknotted genus-9 surface $U \subset S^4$ with $e(U) = -18$, Conway–Galvin construct a form λ_b satisfying [8, Proposition 5.3]

$$Q_{\Sigma_2(U)} \cong (1)^{\oplus 9}, \quad \lambda_b \cong_s \lambda_{\tilde{X}_U}.$$

For the coefficient system $\pi_1(Y_{9, -18}) \rightarrow \mathbb{Z}_2$ sending the fibre to the generator, Lemma 2.10 gives

$$\partial W \cong -Y_{9, -18}, \quad \pi_1(W) \cong \mathbb{Z}_2, \quad H_2(\tilde{W}) \cong \mathbb{Z}_- \oplus \Lambda_2^8, \quad \lambda_{\tilde{W}} \cong \lambda_b.$$

Cap W by $N_{9, -18}$ and retain its zero-section.

$$\begin{array}{ccccc} W & \hookrightarrow & \widehat{W} := W \cup_{Y_{9, -18}} N_{9, -18} & \longleftarrow & N_{9, -18} \\ & & \uparrow & & \\ & & \Sigma_9 & & \end{array}$$

The peripheral and Euler-characteristic calculations give [8]

$$\begin{aligned} \pi_1(\widehat{W}) &= 1, & b_2(\tilde{W}) &= 1 + 2 \cdot 8 = 17, \\ \chi(\tilde{W}) &= 18 = 2\chi(W), & \chi(\widehat{W}) &= 9 + (2 - 9) = 2, \\ H_2(\widehat{W}) &= 0, & w_2(\widehat{W}) &= 0, & \text{ks}(\widehat{W}) &= 0. \end{aligned}$$

Hence

$$\pi_1(\widehat{W}) = 1, \quad H_2(\widehat{W}) = 0, \quad \text{ks}(\widehat{W}) = 0 \xrightarrow{1.21} \widehat{W} \cong S^4$$

by Equation (3). Its zero-section is a $\mathbb{Z}_2$ -surface $S \subset S^4$, and

$$(\lambda_{\tilde{W}})_- \cong 2Q_{\tilde{W}}^{\text{nd}}, \quad (\lambda_b)_- \cong 2b_{\text{ext}}^{\text{nd}}, \quad 9 \not\equiv 4 \pmod{8} \xrightarrow{[8]} Q_{\Sigma_2(S)} \cong b.$$

The branched cover is simply connected because the universal exterior cover is capped along its meridian. Smoothability would therefore give a smooth, simply connected, positive-definite branched cover and

$$E_8 \oplus (1) \cong Q_{\Sigma_2(S)} \cong (1)^{\oplus 9},$$

contradicting Donaldson diagonalisation [24]. Thus S is nonsmoothable. $\square$

The same technology produces the first nontrivial topological knot in S^4 with cyclic knot group [8], answering [2, Questions 4.29(a) and 4.32].

Part 2. Exercises

3. EXERCISES

The exercises from the lectures are listed verbatim; the bracketed reference points to the statement in the notes where each is resolved [17].

Exercise 3.1 (Boundary of a surface exterior, 1.4). Describe the boundary of the exterior of a closed surface in a closed four-manifold.

Exercise 3.2 (Homology and cyclic knot groups, 1.5). Calculate the homology of sphere exteriors in a closed, connected, simply-connected four-manifold X . Deduce the following two statements.

- (a) If $\pi_1(X \setminus S)$ is abelian, then it is cyclic.
- (b) If S additionally has divisibility d , then $\pi_1(X \setminus S) \cong \mathbb{Z}_d$.

Exercise 3.3 (A $\mathbb{Z}_2$ -sphere in $\mathbb{CP}^2$, 1.14). For a $\mathbb{Z}_2$ -sphere $S \subset \mathbb{CP}^2$, calculate the intersection form of the universal cover of the exterior without drawing handle diagrams. Along the way, prove

$$H_1(\partial \tilde{X}_S) \cong \mathbb{Z}_2.$$

Exercise 3.4 (A $\mathbb{Z}_2$ -projective plane in S^4 , 1.15). Repeat Exercise 3.3 for a $\mathbb{Z}_2$ -projective plane $S \subset S^4$. You may use without proof that

$$H_1(\partial \tilde{X}_S) \cong \mathbb{Z}_4.$$

Exercise 3.5 (Relative s -cobordism, 1.19). Suppose X_0 and X_1 are s -cobordant relative to f , and that the relative s -cobordism theorem holds. Prove that

$$f: \partial X_0 \longrightarrow \partial X_1$$

extends to a homeomorphism $X_0 \rightarrow X_1$.

Exercise 3.6 (The equivariant duality isomorphism, 2.2). Describe the isomorphism used in Definition 2.1,

$$H_2(\tilde{X}_S, \partial \tilde{X}_S) \cong \text{Hom}_{\mathbb{Z}[\mathbb{Z}_d]}(H_2(\tilde{X}_S), \mathbb{Z}[\mathbb{Z}_d]).$$

Exercise 3.7 (Equivalence preserves the boundary data, 2.3). Let $S_0, S_1 \subset X$ be equivalent $\mathbb{Z}_d$ -surfaces. Prove that there are an isometry and a bundle isomorphism

$$F: \lambda_{\tilde{X}_{S_0}} \cong \lambda_{\tilde{X}_{S_1}}, \quad H: \bar{v}(S_0) \longrightarrow \bar{v}(S_1),$$

such that

$$\partial F = (H|_{\partial})_*: H_1(\partial \tilde{X}_{S_0}) \longrightarrow H_1(\partial \tilde{X}_{S_1}).$$

Exercise 3.8 (Lawson's unknotting theorem, 1.7). Use Theorem 2.4 to prove Lawson's theorem that $\mathbb{Z}_2$ -projective planes in S^4 are unknotted. You may use without proof that for a $\mathbb{Z}_2$ -surface $S \subset S^4$,

$$\mathbb{Z}_2 = \langle T \mid T^2 = 1 \rangle, \quad Tx = -x \quad \text{for every } x \in H_2(\tilde{X}_S).$$

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